EXTENSIVE MATHEMATICAL DERIVATIONS

Brownian Bridge Diffusion Model

ELBO and the Training Objective

Step-by-step proofs, probability identities,
Gaussian calculations, and mathematical interpretation

PART 3 — VARIATIONAL OBJECTIVE AND GAUSSIAN KL

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1 Conditional generative model

The Brownian Bridge forward process $q(x_{1:T} \mid x_0, y)$ is fixed and tractable. The model learns a reverse generative chain

$$p_\theta(x_{0:T} \mid y) = p(x_T \mid y) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t, y). \tag{1.1}$$

Here $p(x_T \mid y)$ is the terminal prior and $p_\theta(x_{t-1} \mid x_t, y)$ is the learned reverse transition. The learning objective is to maximize the conditional data likelihood

$$\log p_\theta(x_0 \mid y). \tag{1.2}$$

2 Deriving the variational lower bound

Marginalizing the latent trajectory gives

$$p_\theta(x_0 \mid y) = \int p_\theta(x_{0:T} \mid y)\, \mathrm{d}x_{1:T}. \tag{2.1}$$

Introduce the forward distribution as a variational distribution by multiplying and dividing the integrand by $q(x_{1:T} \mid x_0, y)$:

$$p_\theta(x_0 \mid y) = \int q(x_{1:T} \mid x_0, y) \frac{p_\theta(x_{0:T} \mid y)}{q(x_{1:T} \mid x_0, y)}\, \mathrm{d}x_{1:T}.$$

The integral is an expectation under q:

$$p_\theta(x_0 \mid y) = \mathbb{E}_{q(x_{1:T}|x_0,y)} \left[\frac{p_\theta(x_{0:T} \mid y)}{q(x_{1:T} \mid x_0, y)} \right]. \tag{2.2}$$

Taking logarithms,

$$\log p_\theta(x_0 \mid y) = \log \mathbb{E}_q \left[\frac{p_\theta(x_{0:T} \mid y)}{q(x_{1:T} \mid x_0, y)} \right]. \tag{2.3}$$

The logarithm is concave, so Jensen's inequality states

$$\log \mathbb{E}[X] \geq \mathbb{E}[\log X]$$

for positive X. Applying it to (2.3),

$$\log p_\theta(x_0 \mid y) \geq \mathbb{E}_q \left[\log p_\theta(x_{0:T} \mid y) - \log q(x_{1:T} \mid x_0, y) \right].$$

Define the right-hand side as the evidence lower bound:

$$\boxed{\mathcal{L}_{\mathrm{ELBO}} = \mathbb{E}_q \left[\log p_\theta(x_{0:T} \mid y) - \log q(x_{1:T} \mid x_0, y) \right].} \tag{2.4}$$

Maximizing this quantity maximizes a lower bound on the conditional log-likelihood.

3 Expand the forward and reverse factorizations

The forward Brownian Bridge is Markov, so

$$q(x_{1:T} \mid x_0, y) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}, y). \tag{3.1}$$

The reverse generative model is

$$p_\theta(x_{0:T} \mid y) = p(x_T \mid y) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t, y). \tag{3.2}$$

Taking logarithms of the products turns them into sums:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = \mathbb{E}_q \Bigg[& \log p(x_T \mid y) + \sum_{t=1}^{T} \log p_\theta(x_{t-1} \mid x_t, y) \\ & - \sum_{t=1}^{T} \log q(x_t \mid x_{t-1}, y) \Bigg]. \end{aligned} \tag{3.3}$$

This is the expanded ELBO. To expose the KL terms, it is useful to rewrite the forward-chain log density in reverse conditional form.

4 Rearrangement into KL divergences

The joint forward distribution admits the identity

$$q(x_{1:T} \mid x_0, y) = q(x_T \mid x_0, y) \prod_{t=2}^{T} q(x_{t-1} \mid x_t, x_0, y). \tag{4.1}$$

This is simply the chain rule applied in reverse order to the same joint distribution. Substituting (4.1) and (3.2) into (2.4),

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} = \mathbb{E}_q \Bigg[& \log p(x_T \mid y) + \sum_{t=1}^{T} \log p_\theta(x_{t-1} \mid x_t, y) \\ & - \log q(x_T \mid x_0, y) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0, y) \Bigg]. \end{aligned} \tag{4.2}$$

Group the terminal terms:

$$\mathbb{E}_q \left[\log \frac{p(x_T \mid y)}{q(x_T \mid x_0, y)} \right] = -D_{\text{KL}}(q(x_T \mid x_0, y) \,\|\, p(x_T \mid y)) \,. \tag{4.3}$$

For each $t = 2, \ldots, T$, group the matching reverse and posterior terms:

$$\begin{aligned} & \mathbb{E}_{q(x_t \mid x_0, y)} \left[\log p_\theta(x_{t-1} \mid x_t, y) - \log q(x_{t-1} \mid x_t, x_0, y) \right] \\ & \quad = -\mathbb{E}_{q(x_t \mid x_0, y)} \left[D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0, y) \,\|\, p_\theta(x_{t-1} \mid x_t, y)) \right] . \end{aligned} \tag{4.4}$$

The remaining $t = 1$ term is the reconstruction/endpoint term

$$\mathbb{E}_q[\log p_\theta(x_0 \mid x_1, y)]. \tag{4.5}$$

Combining (4.3)–(4.5),

$$\boxed{\begin{aligned}\mathcal{L}_{\text{ELBO}} = & -D_{\text{KL}}(q(x_T \mid x_0, y) \,\|\, p(x_T \mid y)) \\ & -\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0,y)} \left[D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0, y) \,\|\, p_\theta(x_{t-1} \mid x_t, y))\right] \\ & +\mathbb{E}_q[\log p_\theta(x_0 \mid x_1, y)].\end{aligned}} \tag{4.6}$$

Therefore, minimizing the negative ELBO is equivalent to minimizing

$$\boxed{\begin{aligned}\mathcal{J}_{\text{ELBO}} = & D_{\text{KL}}(q(x_T \mid x_0, y) \,\|\, p(x_T \mid y)) \\ & +\sum_{t=2}^{T} \mathbb{E}_q \left[D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0, y) \,\|\, p_\theta(x_{t-1} \mid x_t, y))\right] \\ & -\mathbb{E}_q[\log p_\theta(x_0 \mid x_1, y)].\end{aligned}} \tag{4.7}$$

The first term measures the mismatch between the terminal forward marginal and the chosen prior. The middle terms train each learned reverse transition to match the analytically known Brownian Bridge posterior. The final term handles reconstruction of the endpoint x_0 from x_1.

5 Why the terminal KL term can be omitted during optimization

Consider

$$D_{\text{KL}}(q(x_T \mid x_0, y) \,\|\, p(x_T \mid y)) \,.$$

If the forward schedule and the terminal prior are fixed and contain no trainable parameters θ, this term has zero gradient with respect to θ:

$$\boxed{\nabla_\theta D_{\text{KL}}(q(x_T \mid x_0, y) \,\|\, p(x_T \mid y)) = 0.} \tag{5.1}$$

It can therefore be omitted when optimizing the neural-network parameters. This does not mean the KL term is mathematically zero; it means it is constant with respect to the optimization variables.

6 From the posterior KL to a squared mean error

The analytical posterior derived in Part 2 is

$$q(x_{t-1} \mid x_t, x_0, y) = \mathcal{N}(\widetilde{\mu}_t, \widetilde{\delta}_t I). \tag{6.1}$$

Suppose the learned reverse transition uses the same covariance:

$$p_\theta(x_{t-1} \mid x_t, y) = \mathcal{N}(\mu_\theta(x_t, y, t), \widetilde{\delta}_t I). \tag{6.2}$$

For two d-dimensional Gaussians with the same positive-definite covariance Σ,

$$P = \mathcal{N}(\mu_1, \Sigma), \qquad Q = \mathcal{N}(\mu_2, \Sigma),$$

the KL divergence is

$$\boxed{D_{\mathrm{KL}}(P\|Q) = \frac{1}{2}(\mu_1 - \mu_2)^\top \Sigma^{-1}(\mu_1 - \mu_2).} \tag{6.3}$$

For completeness, this follows by writing the log-density ratio:

$$\log \frac{p(x)}{q(x)} = -\frac{1}{2}(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^\top \Sigma^{-1}(x-\mu_2).$$

Take expectation under P, where $\mathbb{E}_P[x - \mu_1] = 0$ and $\mathbb{E}_P[(x-\mu_1)(x-\mu_1)^\top] = \Sigma$. The equal covariance terms cancel, leaving (6.3).

In our case $\Sigma = \widetilde{\delta}_t I$, so

$$\boxed{D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0, y) \,\|\, p_\theta(x_{t-1} \mid x_t, y)) = \frac{1}{2\widetilde{\delta}_t}\|\widetilde{\mu}_t - \mu_\theta(x_t, y, t)\|^2.} \tag{6.4}$$

Thus the reverse-step KL is a weighted squared error between the true posterior mean and the learned posterior mean.

7 Why noise prediction gives a practical regression objective

The posterior mean can be expressed as a deterministic function of x_t, y, t plus a known scalar coefficient times the forward noise. In the notation of Part 2, one may write

$$\widetilde{\mu}_t = A_t(x_t, y) + c_t \epsilon,$$

where A_t and c_t are determined by the bridge schedule and the chosen noise convention. Parameterize the learned mean by

$$\mu_\theta(x_t, y, t) = A_t(x_t, y) + c_t \epsilon_\theta(x_t, y, t).$$

Subtract:

$$\widetilde{\mu}_t - \mu_\theta = c_t(\epsilon - \epsilon_\theta(x_t, y, t)).$$

Substitute into (6.4):

$$\boxed{D_{\mathrm{KL}} = \frac{c_t^2}{2\widetilde{\delta}_t}\|\epsilon - \epsilon_\theta(x_t, y, t)\|^2.} \tag{7.1}$$

The timestep-dependent factor

$$w_t = \frac{c_t^2}{2\widetilde{\delta}_t}$$

is a positive weighting. Summing/averaging over timesteps and data gives the generic noise-prediction objective

$$\boxed{\mathcal{L}_{\text{noise}} = \mathbb{E}_{x_0,y,t,\epsilon}\left[w_t \|\epsilon - \epsilon_\theta(x_t, y, t)\|^2\right].} \tag{7.2}$$

The exact coefficient c_t, and whether the network predicts standardized noise ϵ or a scaled residual, depends on the parameterization used in the paper. These conventions must not be mixed. The derivation above shows the general route from the reverse KL to a weighted regression loss.

8 Interpretation

The ELBO derivation explains the probabilistic objective; the Gaussian KL calculation explains why it becomes a mean-matching loss; and the reparameterization explains why a network can learn a noise-like quantity without observing x_0 at inference. These are connected but distinct steps:

1. Jensen's inequality produces a lower bound on conditional likelihood.
2. Markov factorization decomposes that bound into terminal, reverse-step, and reconstruction terms.
3. Fixed terminal terms can be dropped from the gradient calculation.
4. Equal-covariance Gaussian KL reduces to a squared distance between means.
5. Noise parameterization converts mean matching into a weighted noise-prediction loss.